

CSC-355 Homework #4

LL(1) grammars and the parsing algorithm

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This homework is an opportunity for you to practice building parse trees, given an (E)BNF that describes a context free language.

Problem 1: LL(1) Grammar for Addition-Chain

- (a) Demonstrate how we would change the following grammar into an LL(1) grammar.

Hint: Use the first grammar transform.

Start $\rightarrow$ Start "+" Num | Num

Num $\rightarrow$ "0" | "1" | "2" | "3" | "4"

- (b) Demonstrate how we would change the following grammar into an LL(1) grammar.

Hint: Use the second grammar transform.

Start $\rightarrow$ Num | Num Plus

Plus $\rightarrow$ "+" Start

Num $\rightarrow$ "0" | "1" | "2" | "3" | "4"

Problem 2: Recognizing List-Manipulation language

Demonstrate how the LL(1) parsing algorithm would operate. The grammar and parse table follow:

- (1) Start $\rightarrow$ Listhead ExprTail
- (2) Exprtail $\rightarrow \epsilon$
- (3) Exprtail $\rightarrow$ "++" Start
- (4) Listhead $\rightarrow$ Elem Listtail
- (5) Listhead $\rightarrow$ ""
- (6) Listtail $\rightarrow$ "::" Listhead
- (7) Elem $\rightarrow$ "a"

	::	++	""	"a"	\$
Start			(1)	(1)	
Exprtail		(3)			(2)
Listhead			(5)	(4)	
Listtail	(6)				
Elem				(7)	

Here are the strings:

- (a) a :: a :: "" ++ a :: a :: ""
- (b) a :: a ++ a :: a :: ""
- (c) a :: "" ++ "" ++ a :: ""
- (d) a :: "" ++ a :: ""